

Algebra Factoring Patterns + Unit Circle Reference

Common Factoring Patterns:

- 1) Greatest Common Factor: $ax + ay = a(x + y)$
- 2) Difference of Squares: $a^2 - b^2 = (a - b)(a + b)$
- 3) Perfect Square Trinomial: $a^2 + 2ab + b^2 = (a + b)^2$
- 4) Perfect Square Trinomial: $a^2 - 2ab + b^2 = (a - b)^2$
- 5) Trinomial ($x^2 + bx + c$): $x^2 + bx + c = (x + m)(x + n)$, $m+n=b$, $mn=c$
- 6) AC Method ($ax^2 + bx + c$): split middle term, then factor by grouping
- 7) Grouping: $ax + ay + bx + by = (a + b)(x + y)$
- 8) Sum of Cubes: $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$
- 9) Difference of Cubes: $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$

Unit Circle Reference (exact values):

Angle	Radians	(cos, sin)	tan
0 deg	0	(1, 0)	0
30 deg	$\pi/6$	$(\sqrt{3}/2, 1/2)$	$\sqrt{3}/3$
45 deg	$\pi/4$	$(\sqrt{2}/2, \sqrt{2}/2)$	1
60 deg	$\pi/3$	$(1/2, \sqrt{3}/2)$	$\sqrt{3}$
90 deg	$\pi/2$	(0, 1)	undefined
120 deg	$2\pi/3$	$(-1/2, \sqrt{3}/2)$	$-\sqrt{3}$
135 deg	$3\pi/4$	$(-\sqrt{2}/2, \sqrt{2}/2)$	-1
150 deg	$5\pi/6$	$(-\sqrt{3}/2, 1/2)$	$-\sqrt{3}/3$
180 deg	π	(-1, 0)	0
210 deg	$7\pi/6$	$(-\sqrt{3}/2, -1/2)$	$\sqrt{3}/3$
225 deg	$5\pi/4$	$(-\sqrt{2}/2, -\sqrt{2}/2)$	1
240 deg	$4\pi/3$	$(-1/2, -\sqrt{3}/2)$	$\sqrt{3}$
270 deg	$3\pi/2$	(0, -1)	undefined
300 deg	$5\pi/3$	$(1/2, -\sqrt{3}/2)$	$-\sqrt{3}$
315 deg	$7\pi/4$	$(\sqrt{2}/2, -\sqrt{2}/2)$	-1
330 deg	$11\pi/6$	$(\sqrt{3}/2, -1/2)$	$-\sqrt{3}/3$
360 deg	2π	(1, 0)	0